Modeling Diffusion with Random Walks: Mesh Parameterization

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Random Walk on $\mathbb{Z}^2$

- **Concept**: Imagine a point starting from 0 on Z walking in both directions (left and right) with equivalent probability $\frac{1}{2}$ every step. If we denote ith step by $X_i \in \{1,-1\}$, then position after n steps is $S_n = \sum_{i=1}^{n} X_i$.
- **Expectation and Mean Squared Displacement**: Since each direction has same probability, the expectation $E[S_n] = 0$. Since each step is independent, the **Mean Squared Displacement** is $E[|S_n|^2] = n$ and $E[|S_n|^2] \propto n$.
- **Distinction**: This means the characteristic displacement of the particle scales proportionally with $\sqrt{n}$, which is a key difference between diffusion and uniform linear motion.
- **Discrete Conservation**: Here we start from 1-Dimension and higher dimension can be concluded similarly. Define $u(x,t)$ as the probability of finding particle at position x at time t. When represented on grid, the probability or status is equal to the average of its neighboring values:

$$u(x,t+\Delta t) = \frac{1}{2}u(x-\Delta x,t) + \frac{1}{2}u(x+\Delta x,t) \tag{1}$$

Connection to Diffusion

In general, random walk is the micro and discrete while diffusion is macro and continuous. To connect them, we apply Taylor Expansion on (1) with the assumption that Δx and Δt are small enough.

- **Left Hand Side**: Expand it to first order on t, we get:

$$u(x,t+\Delta t) \approx u(x,t) + \Delta t\frac{\partial u}{\partial t} \tag{2}$$

- **Right Hand Side**: Expand it to second order on x, we get:

$$\frac{1}{2}u(x-\Delta x,t)+\frac{1}{2}u(x+\Delta x,t) \approx \frac{1}{2}\left[u(x,t) - \Delta x\frac{\partial u}{\partial x} + \frac{(\Delta x)^2}{2}\frac{\partial^2 u}{\partial x^2}\right] + \frac{1}{2}\left[u(x,t) + \Delta x\frac{\partial u}{\partial x} + \frac{(\Delta x)^2}{2}\frac{\partial^2 u}{\partial x^2}\right] \tag{3}$$

So if we add (2) and (3), we get:

$$\frac{\partial u}{\partial t} = \frac{(\Delta x)^2}{2\Delta t}\frac{\partial^2 u}{\partial x^2} \tag{4}$$

Since Δx and Δt are constant, we define **Diffusion Coefficient** $D = \frac{(\Delta x)^2}{2\Delta t}$. If we conclude it into higher dimension, sum of second order partial derivatives is **Laplacian Operator** Δu. Hence, we get:

$$\frac{\partial u}{\partial t} = D\Delta u \tag{5}$$

Visualization of Diffusion

Here is an example of diffusion, the Fundamental Solution of Heat equation, $u(x,t) = \frac{1}{\sqrt{4\pi Dt}}e^{-\frac{x^2}{4Dt}}$

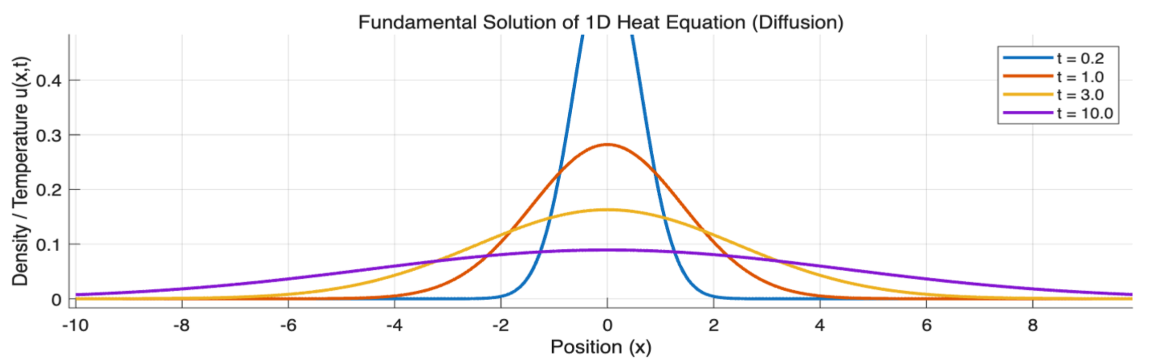

Figure 1. Fundamental Solution of Heat Solution

Representing Diffusion via Matrices

We can represent the state as a vector, $\mathbf{u}(t)$, where we discretize the space into grid points $x_1, x_2, ..., x_N$. Using this, we can create a probability distribution over time at time t.

$$\mathbf{u}(t) = \begin{bmatrix} u(x_1,t) \\ u(x_2,t) \\ \vdots \\ u(x_N,t) \end{bmatrix}, \quad \text{where } \mathbb{P}(S_t = j) = u(x_j,t). \tag{6}$$

From (1), this says that each point gets some probability from its neighbors. This can be written as

$$\mathbf{u}(t+\Delta t) = \mathbf{P}\mathbf{u}(t) \tag{7}$$

where **P** is the transition matrix. **P** has the following properties:

$$\begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 & \cdots \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & \cdots \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix} \tag{8}$$

Where each each row sums to 1 and is sparse (with only neighbors intact). Rearranging (2) and including **P**, the discrete Laplacian matrix becomes

$$\mathbf{L} = \frac{D}{(\Delta x)^2}\begin{bmatrix} -1 & 1 & 0 & \cdots \\ 1 & -2 & 1 & \cdots \\ 0 & 1 & 2 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}. \tag{9}$$

in a 1-D environment. Translating this into 2-D results in the final matrix form in two perspectives:

For a Discrete-time random walk: $\mathbf{u}_{t+1} = \mathbf{P}\mathbf{u}_t$

For a Continuous Diffusion: $d\mathbf{u}/dt = D\mathbf{L}\mathbf{u}$

Generalizing Diffusion in Higher Dimensions

Similar to the 1-D environment, we can extend the random walk into higher dimensions on a d-dimension grid. For each point $\mathbf{x} = (x^1, x^2, ..., x^d)$, $\mathbf{x}$ has $2d$ neighboring points. If we let u($\mathbf{x}$, t) denote the probability of finding particle $\mathbf{x}$ at time t, then it generalizes to

$$u(\mathbf{x}, t+\Delta t) = \frac{1}{2d}\sum_{k=1}^{d}[u(\mathbf{x}+\Delta x\mathbf{e}_k, t) + u(x-\Delta x\mathbf{e}_k, t)] \tag{10}$$

in accordance to the discrete conservation rule. Similar to before, we can apply the Taylor Expansion on this formula (10) to get

$$\frac{\delta u}{\delta t} = \frac{(\delta x)^2}{2d\Delta t}\sum_{k=1}^{d}\frac{\delta^2 u}{\delta x_k^2} \tag{11}$$

Now, we recognize from formula (5) that this equation can be simplified. We can now obtain our final form of

$$\frac{\delta u}{\delta t} = D\Delta u, \text{where D} = \frac{(\delta x)^2}{2d\Delta t} \tag{12}$$

Application: Harmonic Parameterization

- **The Goal**
  If we let the diffusion process run to infinity, the system reaches a steady state ($\frac{\partial u}{\partial t} = 0$). This is determined by Laplace's equation: $\Delta f = 0$. In computer graphics, we use this to flatten 3D meshes (like the Stanford Bunny) into 2D planes for texture mapping.
- **The Variational Trap**
  Mathematically, solving $\Delta f = 0$ is equivalent to minimizing the Dirichlet energy:

$$\min_{f:M\to\mathbb{R}^2}\int ||\nabla f||^2$$

  However, directly minimizing this fails. It yields the trivial solution $f(x) \equiv$ const, meaning the entire 3D bunny collapses into a single 2D point (zero energy, geometrically useless).
- **The Solution: Boundary Conditions**
  To prevent collapse, we must apply Dirichlet Boundary Conditions:
  - First, fix the boundary vertices of mesh to a 2D convex polygon.
  - Then we get the Laplace Equation $\Delta f = 0$ in $\frac{M}{\partial M}$, with $f|_{\partial M}$ fixed.
  - Finally, we have a smooth and unfolded 2D parameterization.

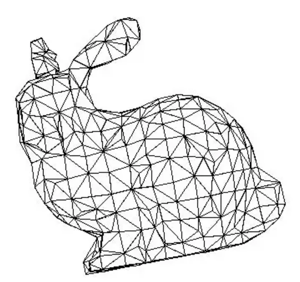

Figure 2. Mesh of Stanford Bunny in 3D

Application: Hodge Decomposition

- **The Theorem**
  Any complex vector field $\vec{v}$ on a manifold can be uniquely decomposed into three orthogonal components:

$$\vec{v} = \nabla f + \nabla \times A + \vec{h}$$

  - ∇f (curl free part): Gradient of a scalar potential.
  - $\nabla \times A$ (divergence free part): Incompressible flow.
  - $\vec{h}$ (harmonic part and $\vec{h} \equiv \vec{0}$ if surface has no holes): Global flows determined entirely by the manifold's topology.
- **Poisson Equation to the Rescue**
  Applying the calculus of variations, the minimizer f must satisfy the Euler-Lagrange equation:

$$\Delta f = \nabla \cdot \vec{v}$$

  By taking the divergence of the known field $\vec{v}$ as a source term, we can solve this Poisson Equation (using our Graph Laplacian matrix L) to find f. Once found, we simply subtract ∇f to isolate the incompressible (divergence free) fluid dynamics.

References

[1] Justin Solomon, Keenan Crane, and Etienne Vouga. Laplace-beltrami: The swiss army knife of geometry processing. Tutorial at the Symposium on Geometry Processing (SGP), July 2014. Presentation slides.